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Deep Learning based Optimal Power Allocation in Underlay D2D Communication

Presentation submitted for the partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Electronics and Communication Engineering

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Outline

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  - Our model comprises of 2 FC parts, 1 convolution part between the two FC parts, and 1 sigmoid part at the end of network.
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  - Deep Neural Network model defined using TensorFlow.
- ***RESULTS***
  - The results of our proposed scheme is exploiting the trade-off between the SE of the DUE and the interference at the CUE.
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  - The spectral efficiency of D2D user equipment and Interference of cellular user equipment are considered.
- ***REFERENCES***
  - The research papers and references we referred to.

Introduction

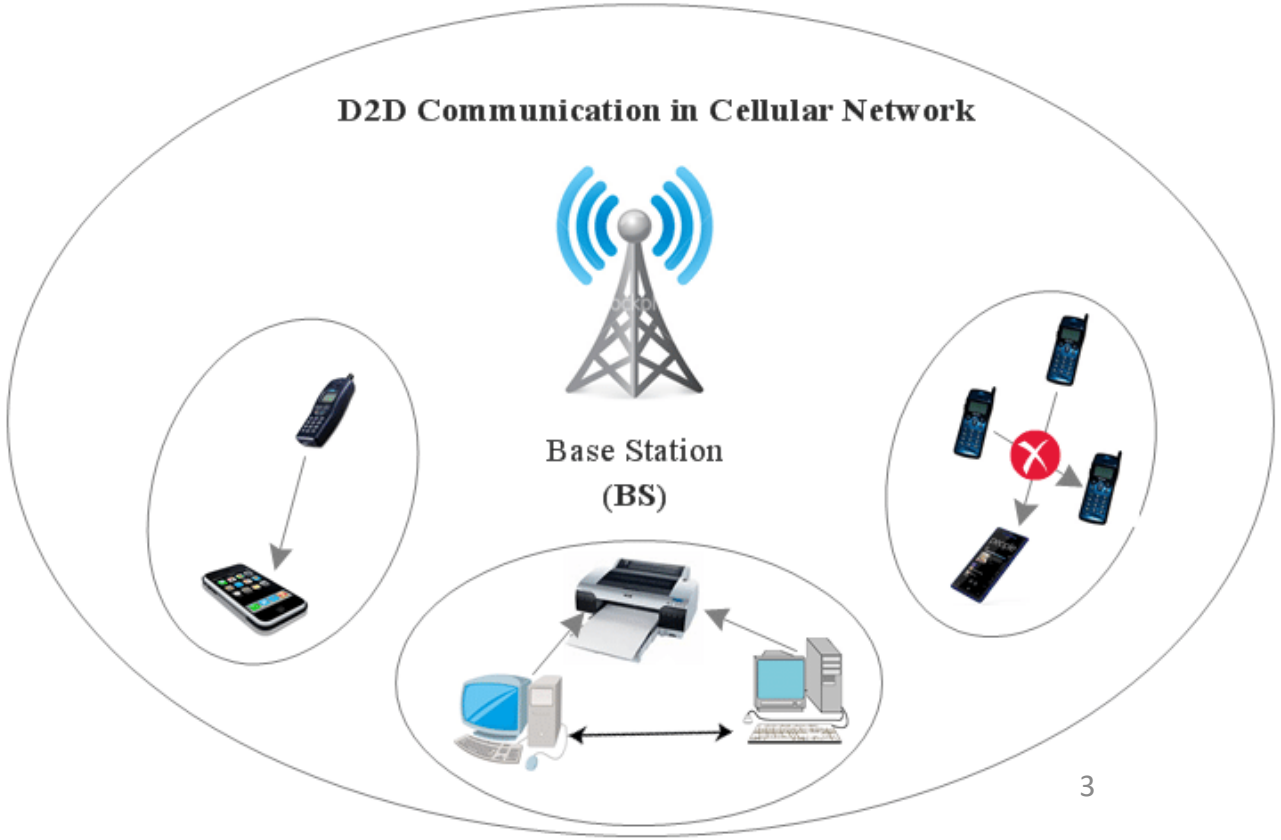
- For challenges of current Cellular networks, Device-To- Device (D2D) communication may be one of key solutions for the next generation (5G) networks.

- D2D communication in cellular networks allows direct communication between two devices without any assistance from Base Station (BS) or infrastructure based networks.

- The technique Deep Neural Networks (DNN), has recently been extensively studied in various fields, including image classification and speech recognition.

- It is well suited to real-time operations because attained DNN model can be used with a low computation time.

- Due to its superior performance compared with conventional approaches. Although the training of a DNN may take some time.



State of the Art (Literature survey)

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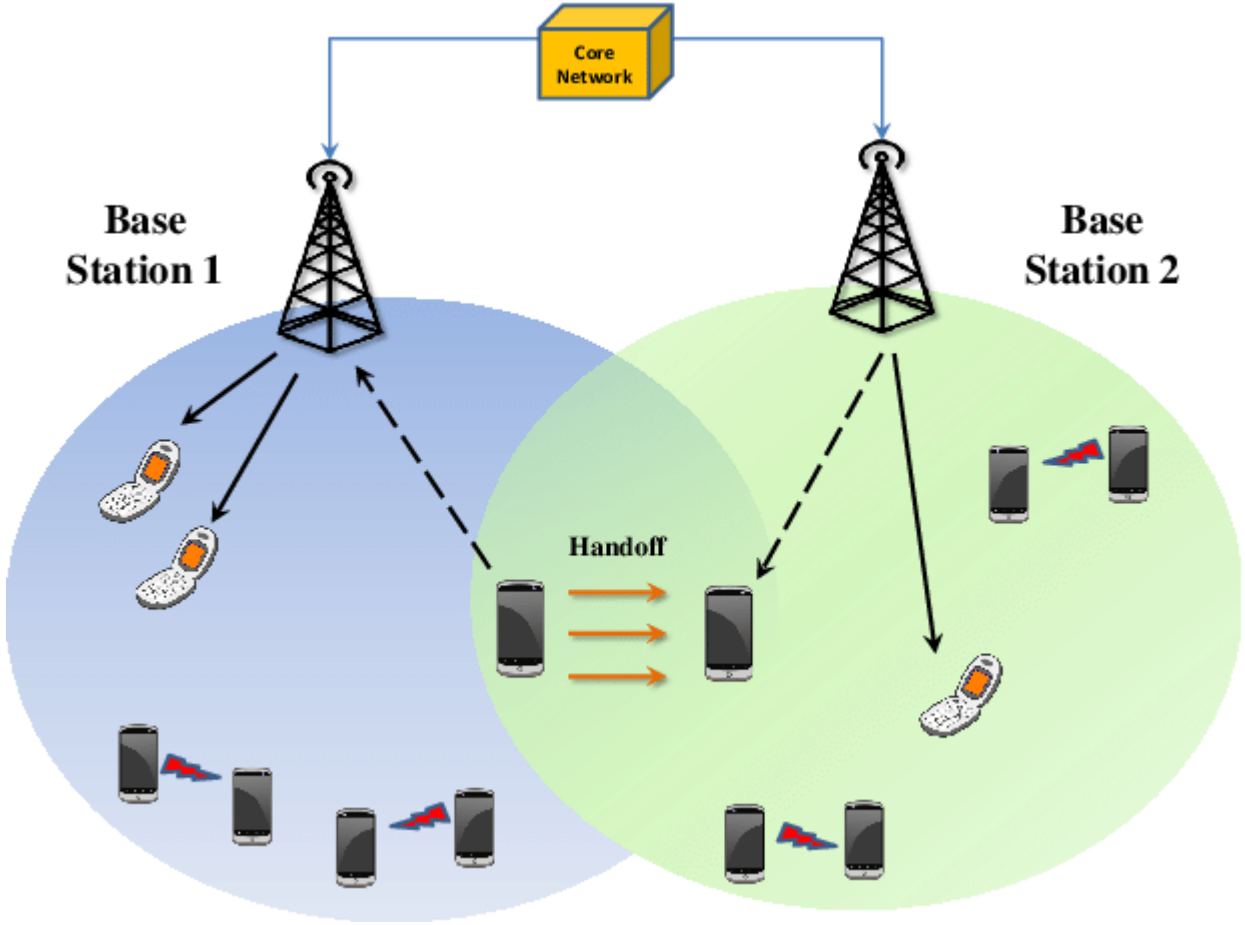
S. no.	Reference work	Analysis
1	Lee, W.; Kim, M.; Cho, D.-H. "Transmit power control using deep neural network for underlay device-to-device communication," IEEE Wirel. Commun. Lett. 2019, 8, 141–144, August 2018.	A transmit power control strategy for DUE is found with the aid of a newly proposed DNN structure. Both the spectral efficiency (SE) of the DUE and the amount of interference at the CUE are considered, such that the SE of the DUE can be improved while alleviating any deterioration in the cellular transmission.
2	D. Feng, L. Lu, Y. Yuan-Wu, G. Y. Li, S. Li, and G. Feng, "Device-to-device communications in cellular networks," IEEE Commun. Mag., vol. 52, no. 4,pp. 49–55, Apr. 2014.	The major issue faced in underlay D2D communication is the power control and interference management between D2D and cellular users. D2D communication in cellular networks is immature and there are still numerous open issues such as interference management, power control.
3	N. Lee, X. Lin, J. G. Andrews, and R. W. Heath, "Power control for D2D underlaid cellular networks: Modelling, algorithms, and analysis," IEEE J. Sel. Areas Commun., vol. 33, no. 1, pp. 1–13, Jan. 2015.	Centralized power control approach leads to improve the cellular network throughput performance due to the additional underlaid D2D links ,distributed power control approach is not enough to guarantee reliable cellular links; however, improves throughput by allowing D2D links to be underlaid.
4	Q. Shi, M. Razaviyayn, Z. Q. Luo, and C. He, "An iteratively weighted MMSE approach to distributed sum-utility maximization for a MIMO interfering broadcast channel," IEEE Trans. Signal Process., vol. 59, no. 9, pp. 4331–4340, Sep. 2011.	A DNN based algorithm for wireless resource allocation DNNs can well-approximate the behaviour of WMMSE, leading to high sum-rate performance and low computational complexity. DNNs have great potential for real-time wireless resource allocation problems. The current work represents a very preliminary step towards understanding the capability of DNNs for this type of problems.

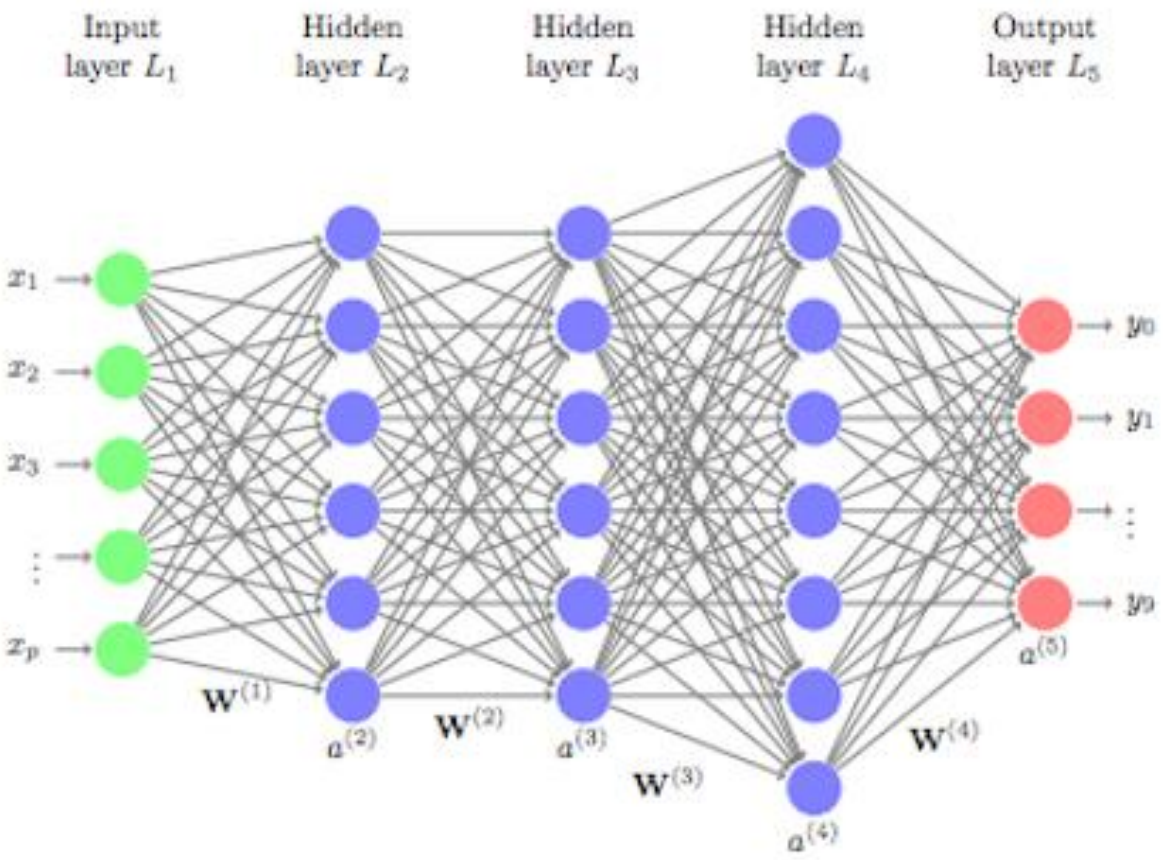
Motivation

- The Optimal Power Allocation in Underlay D2D Communication is of major importance in 5G communication.
- It is generally undertaken by WMMSE approach when using Deep Neural Networks.
- Accounting for cellular interference may allow for improved power allocation.
- As of yet, there has been no systematic investigation of the proposed scheme.
- Successful implementation of this scheme may lead to reduction in better power allocation allowing lower costs for end consumers and a more consistent experience.



Problem Formulation

- We intend to implement DNN scheme for Optimal Power Allocation in Underlay D2D Communication.
- In this the scheme takes the Spectral Efficiency (SE) of the DUE and the amount of interference at the CUE into account.
- We further aim to assess the performance of the proposed scheme and the trade-off between the SE of DUE and CUE interference.
- We compare the performance with conventional schemes and their computation time.



Parameters Used

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- N single-antenna D2D Transceiver Pairs (TPs).
- $\mathbb{I}$ denotes the set of TPs, where the index 0 in $\mathbb{I}$ corresponds to the transmission of the CUE, i.e., Base Station and CUE.
- The transmit power of DUE TP_i is denoted as P_i , where $0 \leq P_i \leq P_M$ and P is the vector of P_i.
- The transmit power of the CUE is fixed at P_{M_C}.
- W and N_0 are the Bandwidth and Noise Spectral Density.
- Channel gain (path loss) and multipath fading is denoted as $h_{i,j}$, where H is the matrix of $h_{i,j}$.
- The Spectral Efficiency of DUE TP_i, $\mathrm{SE}_i = \log_2\left(1 + \frac{h_{i,i}P_i}{N_0 W + \sum_{k\in\mathbb{I}\setminus\{i,0\}} h_{k,i}P_k + h_{0,i}P_{M_C}}\right)$
- The amount of interference caused by the DUE to the CUE, $I_{\mathrm{CUE}} = \sum_{k\in\mathbb{I}\setminus\{0\}} h_{k,0}P_k$
- The SE of the CUE, $\mathrm{SE}_0 = \log_2\left(1 + \frac{h_{0,0}P_{M_C}}{N_0 W + I_{\mathrm{CUE}}}\right)$
- To achieve our goal, we have to follow the optimize condition,

$$\begin{aligned} \underset{P_i}{\text{maximize}} \quad & \sum_{i\in\mathbb{I}\setminus\{0\}} \mathrm{SE}_i \\ \text{s.t.} \quad & 0 \leq P_i \leq P_M, \quad \forall i \in \mathbb{I}\setminus\{0\} \\ & I_{\mathrm{CUE}} \leq I_{\mathrm{THR}}. \end{aligned}$$

System Model

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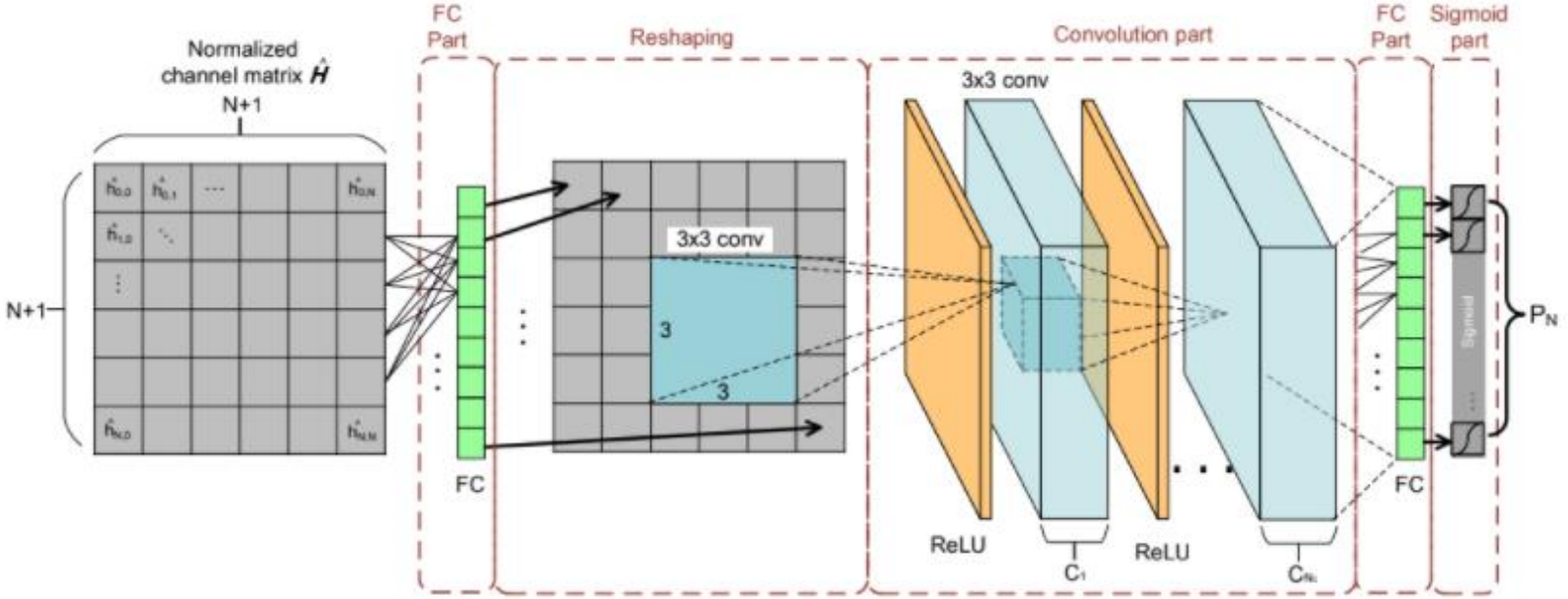
Our model comprises of:-

- 2 FC parts, 1 convolution part between the two FC parts, and 1 sigmoid part at the end of network.
- The role of the first FC part is the fusion of the input data such that all input data can be utilized jointly in the convolution part.
- The convolution part is used to extract meaningful features from input data with low overhead.
- The role of the second FC part is the integration of the outputs from the convolution part.

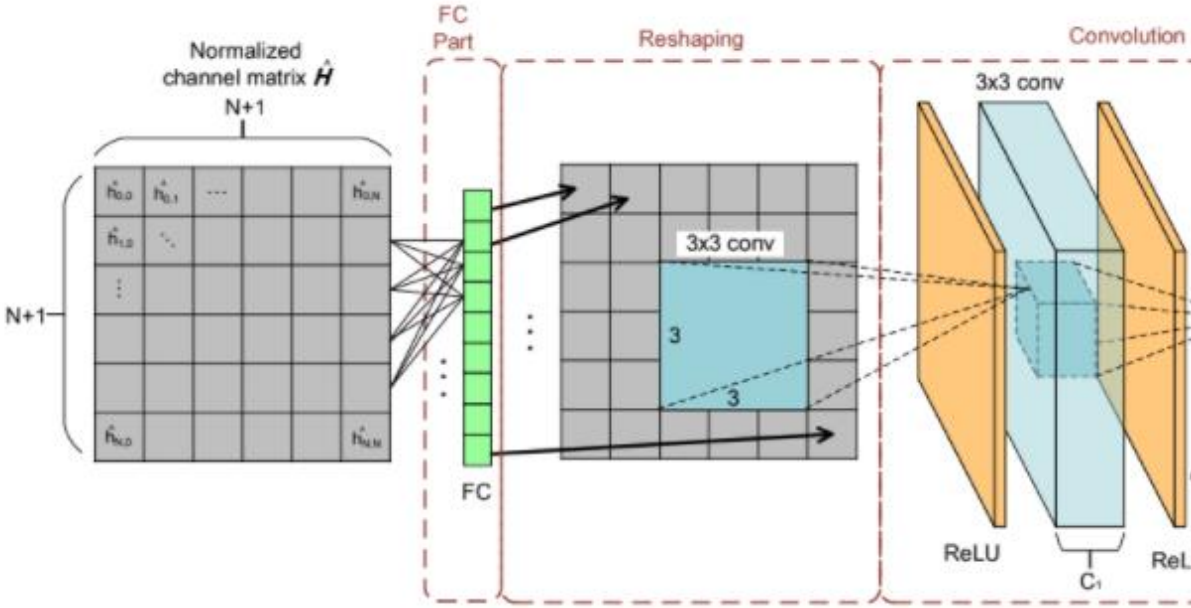


- The sigmoid part is used to restrict the output of the second FC part such that it can be used to model the normalized transmit power.

System Model

- The normalized channel gain $\hat{H}$ becomes the input of the DNN in which, $\hat{h}_{i,j} = \frac{\log_{10}(h_{i,j}) - \mathbb{E}[\log_{10}(h_{i,j})]}{\sqrt{\mathbb{E}[(\log_{10}(h_{i,j}) - \mathbb{E}[\log_{10}(h_{i,j})])^2]}}$.
- $\hat{H}$, is reshaped into a one-dimensional vector, whose length is $(N+1)^2$ and is fed into the first FC part.
- The output of first FC part becomes, $y_{FC2} = W_{FC_1} \mathrm{vec}(\hat{H}) + b_{FC_1}$

  vec(·) denotes the vectorization operation.
- Where W_{FC1}'s shape is $((N+1)^2 \times (N+1)^2)$ and b_{FC1}'s shape is $((N+1)^2 \times 1)$ are the weight and bias of the first FC part.
- Convolution part is composed of N_C sub-blocks.
- Each sub-block comprises:-
  - One convolution layer, which performs a two-dimensional spatial convolution of the input data to extract the spatial features from the data.
  - One rectified linear unit (ReLU) layer, which introduces nonlinearity to the DNN.

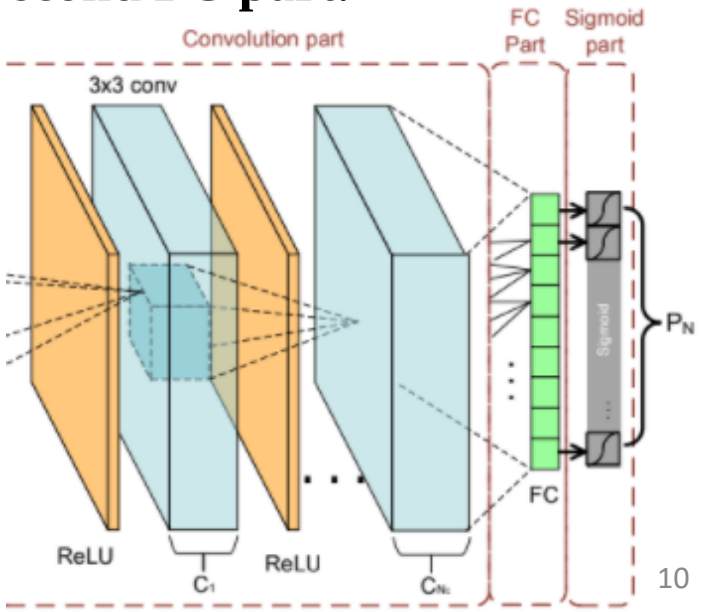


System Model

- We consider 3×3 two-dimensional convolution, $y_C[m, n] = \sum_{i=0}^{2}\sum_{j=0}^{2} x_C[i+m-1, j+n-1]W_C + b_C$,
- Where x_C, W_C, b_C, and y_C are the input, weight, bias, and output of the convolution layer.
- The depth of the convolution layer for the i-th sub-block is set to C_i.
- The output of the second FC part becomes, $y_{FC_2} = W_{FC_2}\mathrm{vec}(x_{FC_2}) + b_{FC}$.

  vec(·) denotes the vectorization operation.
- Where x_{FC2} , y_{FC2} , W_{FC2} , and b_{FC2} are input, output, weight, and bias of the second FC part. respectively.
- The input of the sigmoid part is x_S, the i-th output of the sigmoid part becomes, $\frac{1}{1+e^{-[x_S]_i}}$
- Accordingly, the transmit power of the DUE TP_i is determined as,

$$P_i = \frac{P_M}{1+e^{-[W_{FC_2}\mathrm{vec}(F_{Conv}(W_{FC_1}\mathrm{vec}(\hat{H})+b_{FC_1}))+b_{FC_2}]_i}}$$

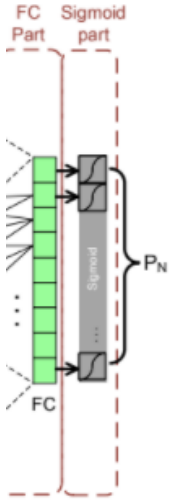


Methodology

- The output of the DNN model is the normalized transmit power P_N, whose value lies between 0 and 1.
- So, the transmit power of the DUE P, is calculated as $P = P_M P_N$.
- To this end, following Loss Function L_T, is considered for the training:

$$L_T = -\lambda \sum_{i \in \mathbb{I} \setminus \{0\}} \mathrm{SE}_i + (1-\lambda) \tanh\left(\frac{[I_{\mathrm{CUE}} - I_{\mathrm{THR}}]^+}{\delta}\right)$$

where $0 \leq \lambda \leq 1$ and $\tanh(\cdot)$ is a hyperbolic tangent



- Where λ determines whether the DNN places greater emphasis on the maximization of the SE of the DUE or on the regularization of the interference at the CUE.
- When λ is close to 1, the transmit power of the DUE is determined without consideration of the interference at the CUE.
- When λ is close to 0, the interference at the CUE is strictly limited while the SE of the DUE is barely considered.

Methodology

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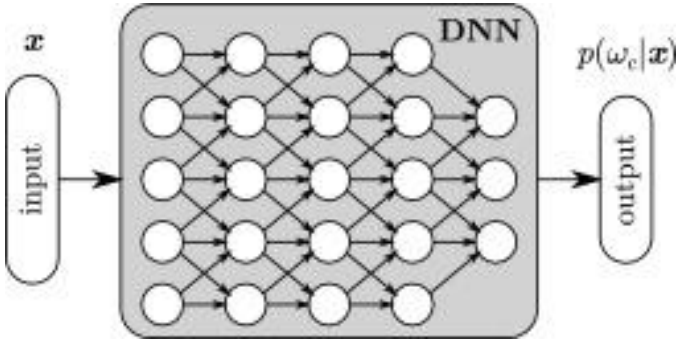
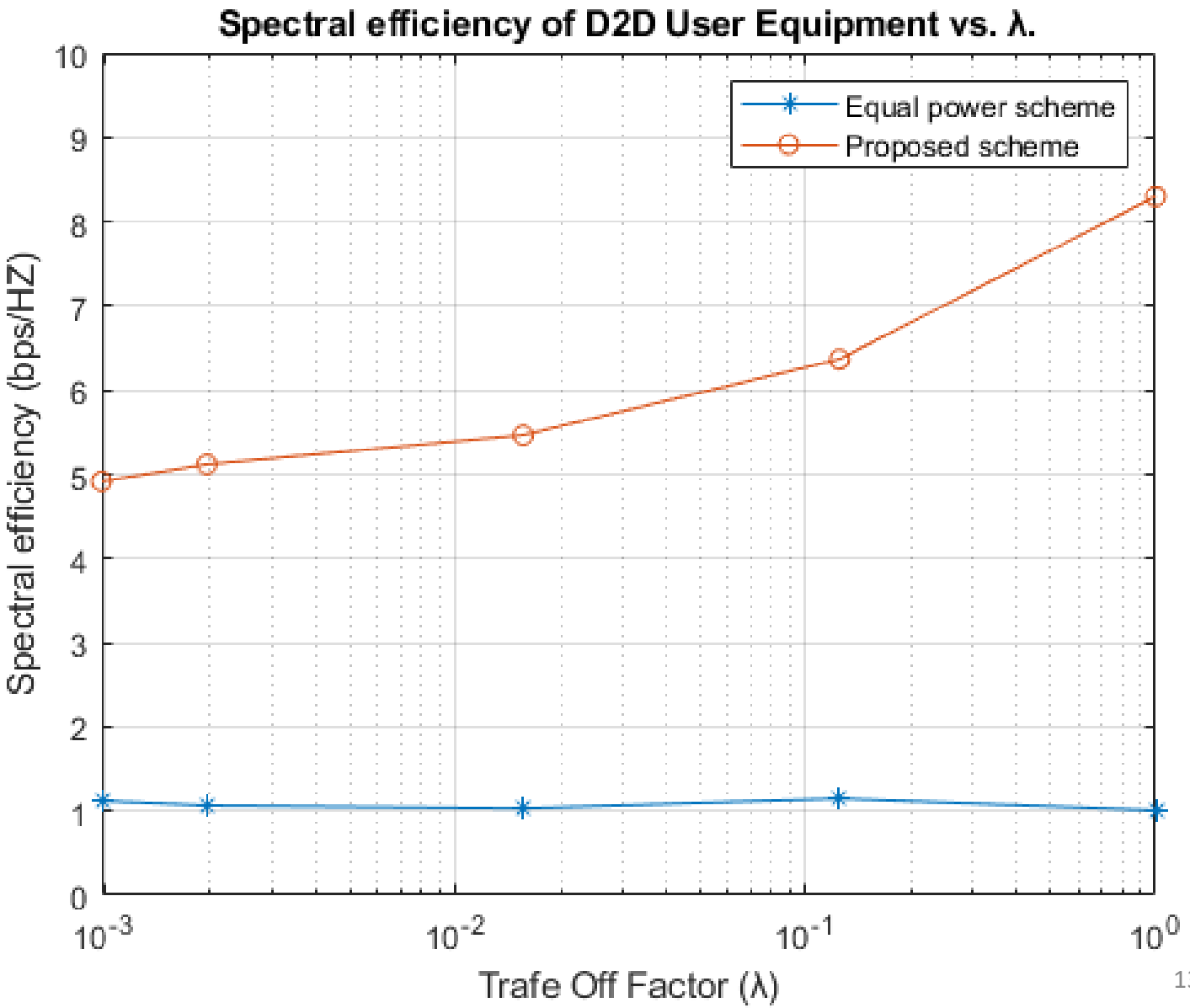
- Scheme used – Deep Neural Network based structure.
- Architecture - 5 layers of hidden layers.
- Weights and biases getting dropout to improve overfitting.



- Our DNN model is defined using TensorFlow because it does parallel computation.
- The parameters like weights and biases are randomly initialized.
- Initialization and training are achieved using Adaptive Moment Estimation (Adam).
- The training (train dataset) and evaluation (test dataset) are taken for 100,000 channel samples.
- Evaluating the performance by calculating SINR at each step.
- Iteration done for different values of Trade off factor (λ) ranging from 0 to 1.
- Generating the final SE and Interference values for comparison against Conventional schemes.

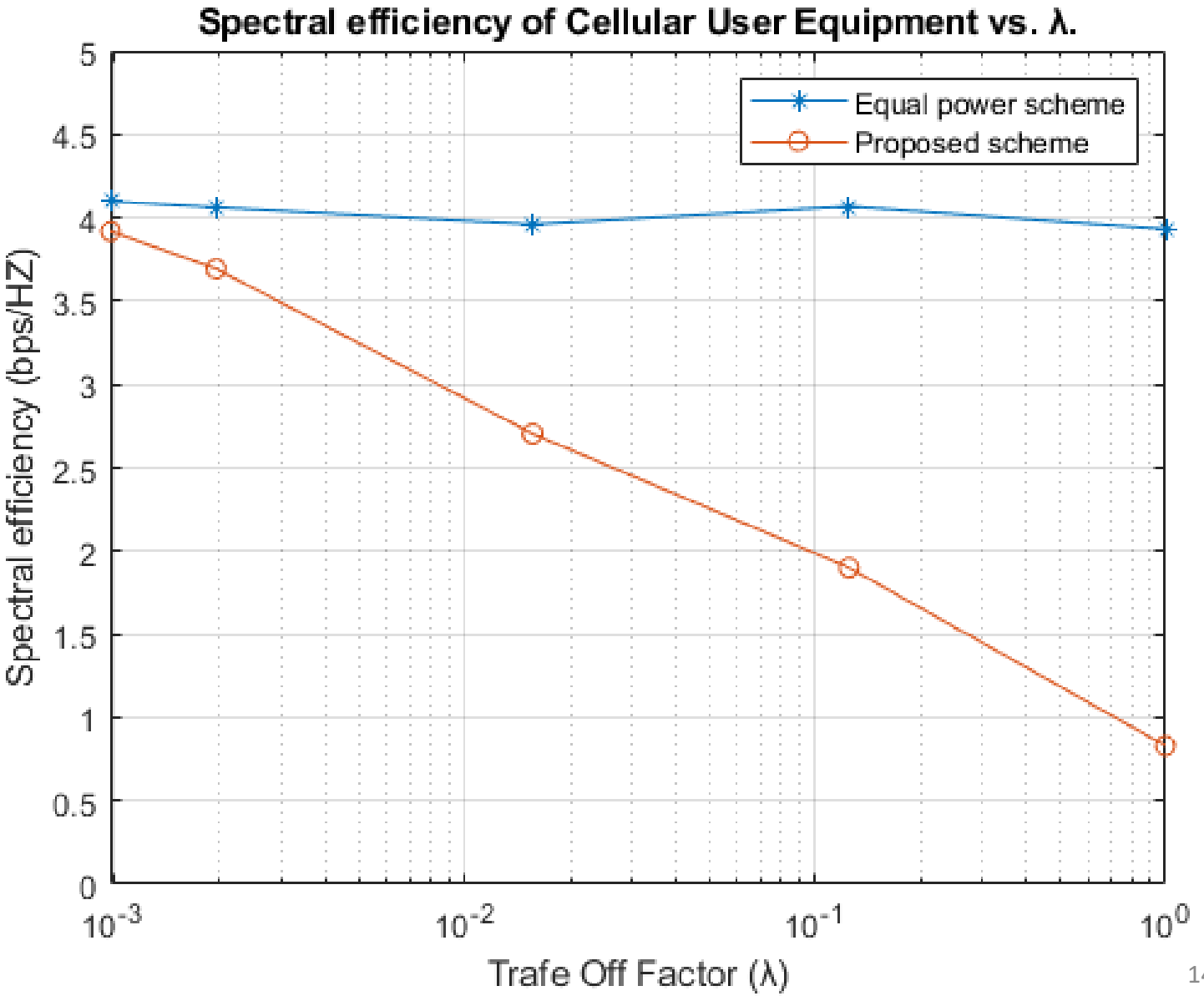
Results

- The Spectral Efficiency of DUE (bps/Hz) of the proposed scheme increases as λ increases.
- It is because the DNN is trained to place more emphasis on the improvement of the Spectral Efficiency of the DUE.
- The Spectral Efficiency of the DUE of conventional scheme i.e., the equal power scheme is not varying.
- Because λ is only related to the DNN through its loss function in training.



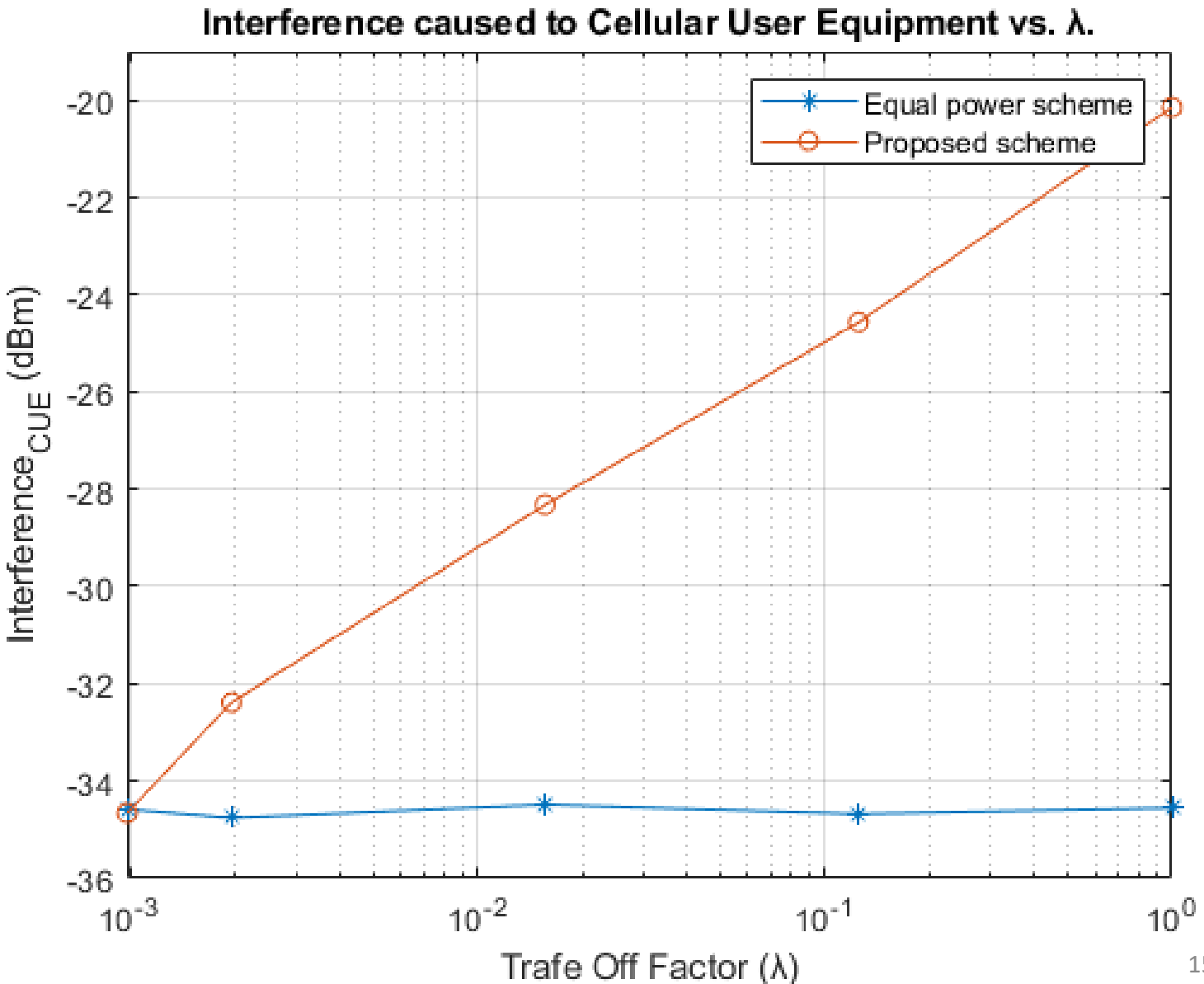
Results

- The Spectral Efficiency of CUE (bps/Hz) of the proposed scheme increases as λ decreases.
- It is because the interference caused to the CUE becomes more important than the SE of the DUE in training.
- The Spectral Efficiency of the CUE of conventional scheme i.e., the equal power scheme is not varying because λ is only related to the DNN through its loss function in training.
- The SE of the CUE of equal power scheme is higher due to less interference caused by the DUE.



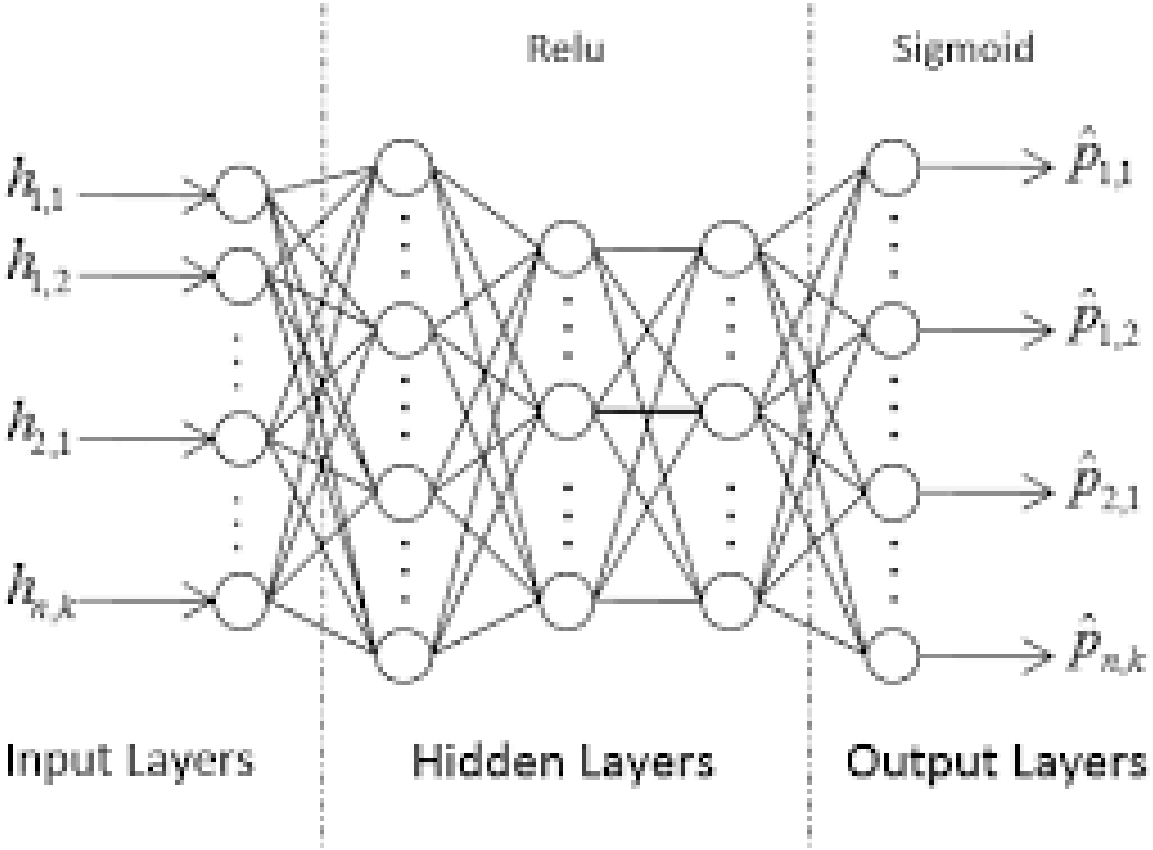
Results

- As λ increases, the more emphasis is on the improvement of the Spectral Efficiency of the DUE.
- As λ increases, the Interference caused to the CUE becomes less important.
- Because of which the Interference at the CUE (dBm) increases in similar fashion.
- In the equal power scheme, the transmit power of all DUE transmitters is set such that the constraint ($I_{CUE} \leq I_{THR}$) is satisfied with equality.



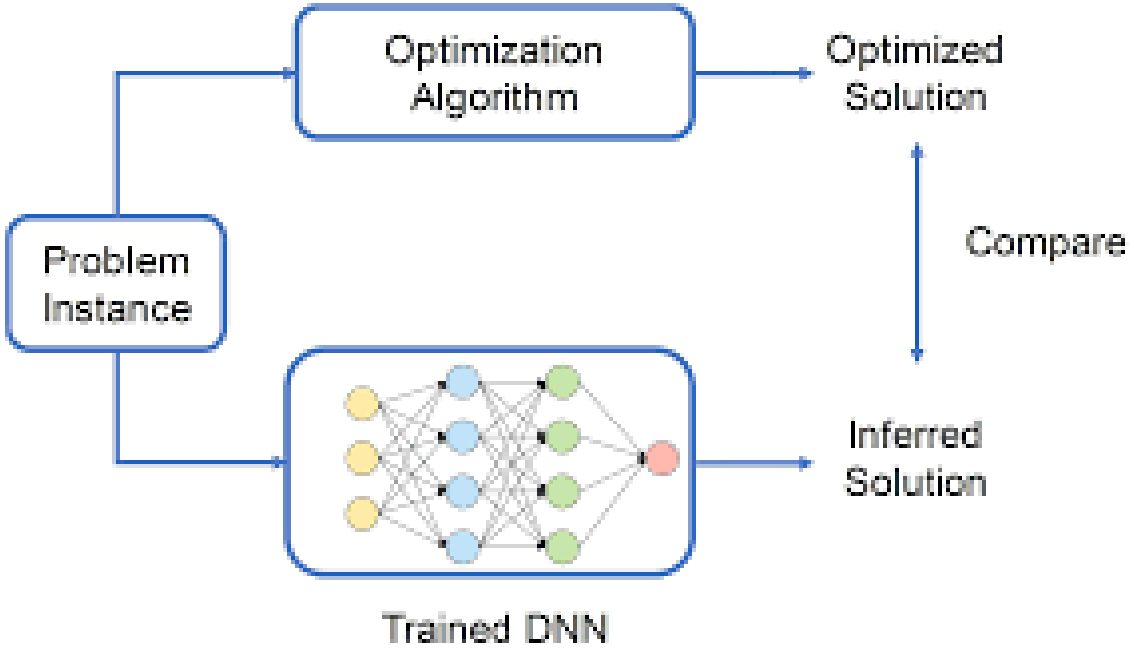
Concluding Results

- **In our Proposed Scheme (DNN), the constraint on I_{CUE} is satisfied when $\lambda \leq 0.001$.**
- **The Spectral Efficiency of the DUE for our proposed scheme at $\lambda = 0.001$ is much higher than that for the Equal Power Scheme.**
- **This confirms the benefit of our proposed scheme.**
- **From these results, we can see that our proposed scheme is exploiting the trade-off between the SE of the DUE and the interference at the CUE.**
- **Also allowing it to achieve higher performance than conventional scheme.**



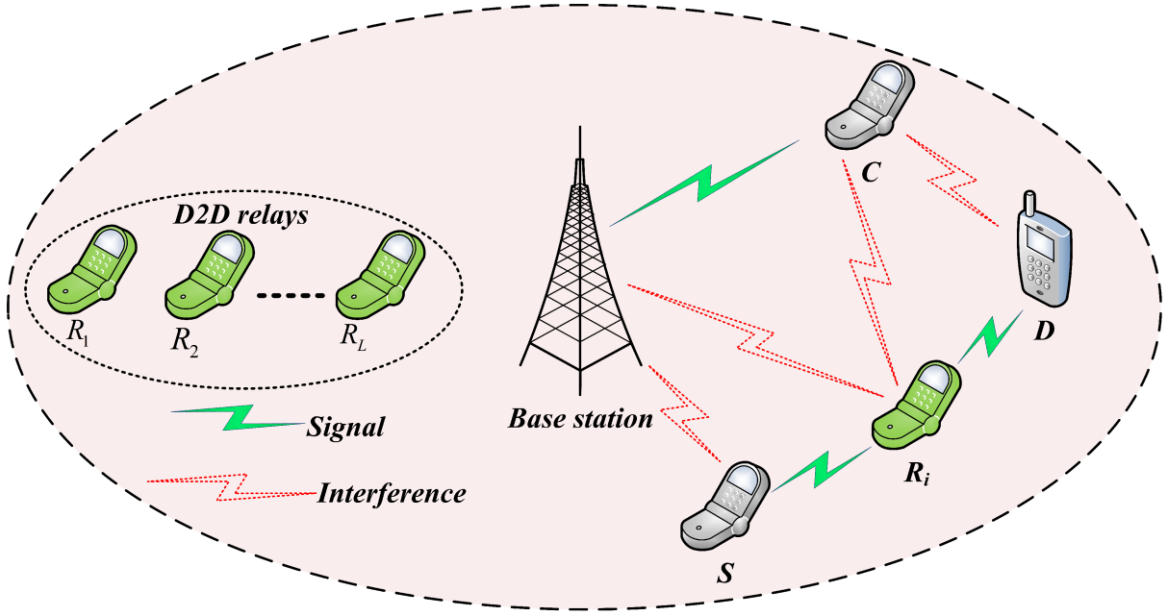
Conclusion

- **Device to Device (D2D) Communication - Direct communication between two mobile users without traversing the Base Station or core network.**
- **Major issues - Power Control and Interference Optimization.**
- **WMMSE scheme - Accounts for the Spectral Efficiency of D2D User Equipment only.**
- **Proposed scheme – DNN based Transmit Power Control.**
- **Both Spectral Efficiency of D2D User Equipment and Interference of Cellular User Equipment are considered.**
- **Results show that the proposed scheme model shows higher performance than conventional scheme.**

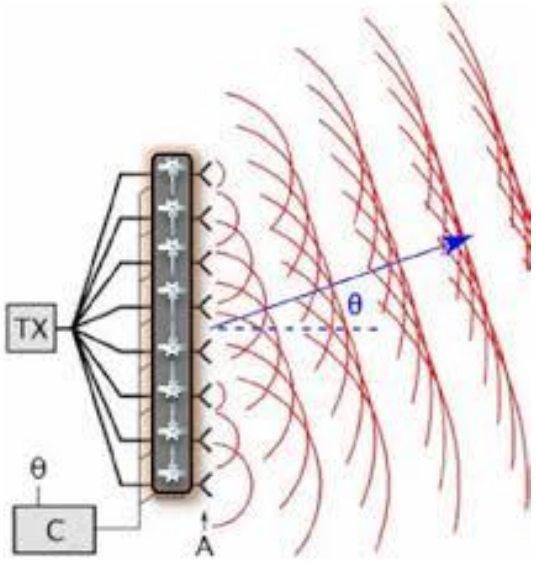


Future Scope

- **Joint beamforming and power allocation.**



- **Relay-Assisted D2D Communications.**



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I want to acknowledge and express my deepest gratitude to our supervisor for continuously guiding us through this BTP work and motivating us to contribute to this research work –

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THANK YOU !!